

The Power of Quantifier Shifts

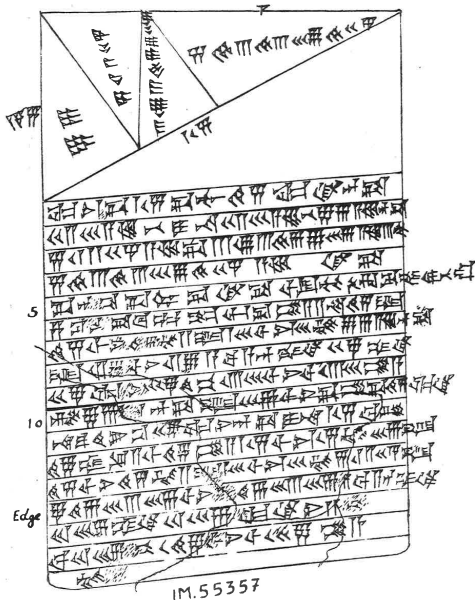
(Dedicated to V.P. Orevkov)

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Hilbert's paradigm

A proof is a sequence of formulas $A_1 \dots A_n$ such that for all i

- ▶ A_i is an instance of an axiom or
- ▶ A_i follows from $A_{j_1} \dots A_{j_k}$ by application of a rule ($j_1, \dots, j_k < i$).
- ▶ A_n is the result of the proof.

$$\forall x \exists y (P(y) \supset P(x)) \quad P(a) \supset P(a)$$

$$\Downarrow$$

$$\forall x P(x) \supset \forall x P(x)$$

$$\exists y \forall x (P(y) \supset P(x)) \quad (P(a) \supset P(b)) \vee (P(b) \supset P(c))$$

- (1) That Kurt Gödel is Austrian entails that Kurt Gödel is Austrian.
- (2) Hence, that Kurt Gödel is Austrian entails that everyone is Austrian.
- (3) That is, if Kurt Gödel is Austrian, then all people are Austrian.
- (4) Therefore, there exists a person such that, if that person is Austrian, then all people are Austrian.

$$\begin{array}{c}
 A(a) \vdash A(a) \\
 \hline
 A(a) \vdash \forall y A(y) \\
 \hline
 \vdash A(a) \rightarrow \forall y A(y) \\
 \hline
 \vdash \exists x (A(x) \rightarrow \forall y A(y))
 \end{array}$$

Side variables

b is a side variable of a in π (written $a <_{\pi} b$) if π contains a strong-quantifier inference of the form

$$\frac{\Gamma \vdash \Delta, A(a, b, \vec{c})}{\Gamma \vdash \Delta, \forall x A(x, b, \vec{c})}$$

or of the form

$$\frac{A(a, b, \vec{c}), \Gamma \vdash \Delta}{\exists x A(x, b, \vec{c}), \Gamma \vdash \Delta}$$

Suitable quantifier inferences

A quantifier inference is suitable for a proof π if either it is a weak-quantifier inference, or the following three conditions are satisfied:

- ▶ (substitutability) the characteristic variable does not appear in the conclusion of π .
- ▶ (side-variable condition) the relation $<_{\pi}$ is acyclic.
- ▶ (weak regularity) the characteristic variable is not the characteristic variable of another strong-quantifier inference in π .

(LK⁺)

The calculus LK⁺ is defined like LK, except that we instead allow all weak and strong quantifier inferences with the proviso that they be suitable for the proof.

Weakly suitable quantifier inference

A quantifier inference is weakly suitable for a proof π if either it is a weak-quantifier inference or it satisfies substitutability, the side-variable condition, and

- ▶ (very weak regularity) whenever the characteristic variable is also the characteristic variable of another strong-quantifier inference in π , then it has the same critical formula.

LK^{++}

The calculus LK^{++} is the extension of LK^+ that results from allowing all weakly suitable quantifier inferences.

LK

$$\begin{array}{c}
 \frac{A(a) \vdash A(a)}{A(a) \vdash A(a), B} \\
 \hline
 \vdash A(a), A(a) \rightarrow B \\
 \hline
 \vdash A(a), \exists x (A(x) \rightarrow B) \\
 \hline
 \vdash \exists x (A(x) \rightarrow B), A(a) \\
 \hline
 \vdash \exists x (A(x) \rightarrow B), \forall x A(x) \quad B \vdash B \\
 \hline
 \forall x A(x) \rightarrow B \vdash \exists x (A(x) \rightarrow B), B \\
 \hline
 \forall x A(x) \rightarrow B, A(b) \vdash \exists x (A(x) \rightarrow B), B \\
 \hline
 \forall x A(x) \rightarrow B \vdash \exists x (A(x) \rightarrow B), A(b) \rightarrow B \\
 \hline
 \forall x A(x) \rightarrow B \vdash \exists x (A(x) \rightarrow B), \exists x (A(x) \rightarrow B) \\
 \hline
 \forall x A(x) \rightarrow B \vdash \exists x (A(x) \rightarrow B)
 \end{array}$$

LK⁺

$$\frac{\frac{\frac{A(a) \vdash A(a)}{A(a) \vdash \forall x A(x)} \quad B \vdash B}{A(a), \forall x A(x) \rightarrow B \vdash B}}{\forall x A(x) \rightarrow B, A(a) \vdash B} \quad \frac{\forall x A(x) \rightarrow B \vdash A(a) \rightarrow B}{\forall x A(x) \rightarrow B \vdash \exists x (A(x) \rightarrow B)}$$

Soundness

Theorem.

If a sequent is LK^{++} -derivable, then it is already LK -derivable.

Proof. Let π be an LK^{++} -proof. Replace every unsound universal quantifier inference by an $\rightarrow L$ inference:

$$\frac{\Gamma \vdash \Delta, A(a) \quad \forall x A(x) \vdash \forall x A(x)}{\Gamma, A(a) \rightarrow \forall x A(x) \vdash \Delta, \forall x A(x)}$$

Similarly replace every unsound existential quantifier by an $\rightarrow L$ inference

$$\frac{\exists x A(x) \vdash \exists x A(x) \quad A(a), \Gamma \vdash \Delta}{\Gamma, \exists x A(x), \exists x A(x) \rightarrow A(a) \vdash \Delta}$$

Cut with formulae of the form

$$\vdash \exists y (A(y) \rightarrow \forall x A(x))$$

or of the form

$$\vdash \exists y (\exists x A(x) \rightarrow A(y)),$$

both of which are easily derivable.

Conditions Are Necessary for Soundness!

1. (substitutability is not fulfilled)

$$\frac{A(a) \vdash A(a)}{A(a) \vdash \forall x A(x)}$$

incorrect result

2. (side-variable condition is not fulfilled)

$$\frac{\frac{\frac{A(a, b) \vdash A(a, b)}{\exists y A(a, y) \vdash A(a, b)} *_1}{\exists y A(a, y) \vdash \forall x A(x, b)} *_2}{\forall x \exists y A(x, y) \vdash \forall x A(x, b)}$$

$$\frac{\forall x \exists y A(x, y) \vdash \forall x A(x, b)}{\forall x \exists y A(x, y) \vdash \exists y \forall x A(x, y)}$$

Note that in $*_1$ and $*_2$ we have $b <_{\pi} a$ and $a <_{\pi} b$, respectively, incorrect result.

3. (very weak regularity is not fulfilled)

$$\frac{\frac{A(a) \vdash A(a)}{\exists x A(x) \vdash A(a)}}{\frac{A(a) \vdash A(a)}{A(a) \vdash \forall x A(x)}} \quad \frac{A(a) \vdash A(a)}{A(a) \vdash \forall x A(x)}$$

$$\frac{\exists x A(x) \vdash A(a)}{\exists x A(x) \vdash \forall x A(x)}$$

incorrect result.

The calculus LK_{shift} is obtained by extending LK with the following rules:

$$\frac{\Gamma, \kappa[Qx A \triangleleft B] \vdash \Delta}{\Gamma, \kappa[Q'x (A \triangleleft B)] \vdash \Delta}$$

$$\frac{\Gamma \vdash \Delta, \kappa[Qx A \triangleleft B]}{\Gamma \vdash \Delta, \kappa[Q'x (A \triangleleft B)]}$$

$$\frac{\Gamma, \kappa[A \triangleleft Qx B] \vdash \Delta}{\Gamma, \kappa[Q'x (A \triangleleft B)] \vdash \Delta}$$

$$\frac{\Gamma \vdash \Delta, \kappa[A \triangleleft Qx B]}{\Gamma \vdash \Delta, \kappa[Q'x (A \triangleleft B)]}$$

where $\kappa[\cdot]$ is a context, $\triangleleft \in \{\wedge, \vee, \rightarrow\}$ and $Q' = Q$, except if $\triangleleft$ is $\rightarrow$ and Q is taken from the antecedent, in which case Q' is opposite. We refer to these rules as *deep quantifier shifts*.

Theorem (cf. Aguilera, Baaz)

Cut-free $\mathbf{LK}^+$ simulates cut-free double-exponentially, i.e., every LK_{shift} -provable sequent is $\mathbf{LK}^+$ -provable and there is a double exponential function that bounds the length of the least cut-free $\mathbf{LK}^+$ -proof of a $\mathbf{LK}^+$ -provable sequent in terms of its least cut-free LK_{shift} -proof.

Remark

Since there are at most n deep quantifier shifts, the size of the resulting proof is bounded by

$$n^{2^n} = 2^{2^n \cdot \log(n)} \approx 2^{2^n}$$

and this is elementary.

Theorem

Use the following modification of Orevkov's worst-case sequence $\{S_i\}_{i < \omega}$

1. the size of S_i is exponentially in i , and S_i contains only weak quantifiers;
2. there is no bound on the size of their smallest cut-free **LK**-proofs that is elementary in i ;
3. the size of these proofs (with cuts), however, is elementary bounded in i ;
4. all cut formulas are closed and prenex.

Theorem

*There is no elementary function bounding the length of the shortest cut-free **LK**-proof of a formula in terms of its shortest cut-free LK_{shift} -proof.*

Proof.

Use of the family of sequents $\{S_i\}_{i < \omega}$, their short cut-free proof π_0 whose end sequent is of the form:

$$A_0 \rightarrow A_0, \dots, A_m \rightarrow A_m, \Gamma_i \vdash \Delta_i. \quad (1)$$

Place the quantifiers pairwise such that the strong quantifiers are inside the weak quantifiers. We have now a cut-free proof with the end-sequent

$$\forall x_0^0 \exists x_1^0 \dots (\hat{A}_0 \rightarrow \hat{A}_0), \dots, \forall x_0^m \exists x_1^m \dots (\hat{A}_m \rightarrow \hat{A}_m), \Gamma_i \vdash \Delta_i. \quad (2)$$



Theorem

Cut-free LK_{shift} proofs of this sequents are short, Cut-free LK -proofs of this sequents are long.

Proof.

Skolemize the end-sequent and calculate the Herbrand sequent(elementary). Replace maximal Skolem terms $f(t, \dots)$ by t , where t belongs to the dual quantifier. This preserves the propositional validity of a Herbrand sequent. Repeat this until the sequents have the propositional forms $\dots, A_i \rightarrow A_i, \dots \Pi \vdash \Gamma$, As these sequents are propositionally valid this sequents $\Pi \vdash \Gamma$ are propositionally valid and therefore large contradiction. $\square$

Note that cut-free LK^{++} - proofs can be reduced to cut-free LK^+ -proofs in an elementary way.

Definition

The introduced Skolem functions in Andrews method do not depend on the weak quantifiers $(Q_1x_1) \dots (Q_nx_n)$ that dominate the strong quantifier (Qx) , but depend on the subset of $\{x_1, \dots, x_n\}$ appearing (free) in the subformula dominated by (Qx) .
Andrews Skolemization may allow shorter Herbrand sequent.

Example

Consider the sequent

$$S: \forall x \forall y (\exists z P(x, z) \vee Q(y, x)) \vdash .$$

Then its structural Skolemization is

$$\forall x \forall y (P(x, f(x, y)) \vee Q(y, x)) \vdash$$

and its Andrews Skolemization is

$$\forall x \forall y (P(x, g(x)) \vee Q(y, x)) \vdash .$$

Remark

Consider

$$\exists y(\exists x A(x) \rightarrow A(y)).$$

For structural Skolemization

$$\exists y(A(f(y)) \rightarrow A(y))$$

the shortest Herbrand disjunction is

$$A(f(f(a)) \rightarrow A(f(a)) \vee A(f(a)) \rightarrow A(a).$$

For Andrews Skolemization

$$\exists y(A(c) \rightarrow A(y))$$

the shortest Herbrand disjunction is

$$A(c) \rightarrow A(c).$$

Remark

Andrews Skolemization has been originally introduced by the equal satisfiability of

$$A(\exists x B(x_1 \dots x_n))$$

and

$$A(P(x_1 \dots x_n)) \wedge \forall x_1 \dots x_n (\exists x B(x_1 \dots x_n) \leftrightarrow P(x_1 \dots x_n))$$

$x_1 \dots x_n$ bound by weak quantifiers outside B and occurring in B , P new predicate. Skolemize the right component with structural Skolemization and use the equivalence to obtain

$$A(B(f(x_1 \dots x_n), x_1 \dots x_n)).$$

Theorem

Cut-free $\mathbf{LK}^+$ and $\mathbf{LK}^{++}$ -proofs can be Skolemized by substitution with Andrews Skolemization in an elementary way but not in structural Skolemization. Structural Skolemization leads to introduce cuts.

Theorem

Cut-free $\mathbf{LK}^{++}$ -proofs of end-sequents with Andrews Skolemization can be deSkolemized in a polynomial way depending on the number of quantifiers in the target end-sequents.

Non-classical Logics

Example

Consider the following $\mathbf{LJ}^+$ and $\mathbf{LJ}^{++}$ -proof:

$$\frac{\frac{A(a) \vdash A(a)}{A(a) \vdash \forall x A(x)} \quad \frac{A(f(a)) \vdash A(f(a))}{\forall x A(x) \vdash A(f(a))}}{A(a) \vdash A(f(a))} \frac{}{\vdash A(a) \rightarrow A(f(a))} \frac{}{\vdash \exists x (A(x) \rightarrow A(f(x)))} \quad (3)$$

If there were a cut-free proof then $\exists x (A(x) \rightarrow A(f(x)))$ would be valid.

So $\mathbf{LJ}^+$ and $\mathbf{LJ}^{++}$ are neither sound for intuitionistic logic nor admit cut-elimination.

Intuitionistically Invalid Classical Shift Principles

- Constant Domain (CD):

$$\forall x(A(x) \vee B) \rightarrow \forall xA(x) \vee B$$

- Independence of Premise (IP)

$$(B \rightarrow \exists xA(x)) \rightarrow \exists x(B \rightarrow A(x))$$

- Quantifier Switch (SW):

$$(\forall xA(x) \rightarrow B) \rightarrow \exists x(A(x) \rightarrow B)$$

Theorem

An intermediate first-order logic admits structural and Andrews Skolemization iff all classical quantifier shifts are available.

Epsilon calculus in its sequent formulation allows for a non-elementary speed-up on cut-free $\mathbf{LK}^+$ and $\mathbf{LK}^{++}$ proofs by its cut-free proofs.

Definition

An epsilon term $\varepsilon_x A(x)$ overbinds if it is generated by the inferences

$$\frac{\Pi \vdash \Delta, A(t)}{\Pi \vdash \Delta, A(\varepsilon_x A(x))} \exists_r$$

$$\frac{A(t), \Pi \vdash \Delta}{A(\varepsilon_x \neg A(x)), \Pi \vdash \Delta} \forall_l$$

when t contains some epsilon-term.

Example

$$\frac{\Pi \vdash \Gamma, \varepsilon_x(\neg x = x) = \varepsilon_x(\neg x = x)}{\Pi \vdash \Gamma, \varepsilon_y(y = \varepsilon_x(\neg x = x)) = \varepsilon_x(\neg x = x)}$$

Theorem

There is a sequence of cut-free $\mathbf{L}_\varepsilon$ -proofs of S_i such that

- 1. the size of S_i is elementary in i ;*
- 2. the end-sequents S_i are translations of first-order sequents S'_i with weak quantifiers only;*
- 3. the size of these $\mathbf{L}_\varepsilon$ -proofs is elementarily bounded in i ;*
- 4. There is no elementary bound on the size of the smallest cut-free $\mathbf{LK}^{++}$ -proofs of S'_i .*

Theorem

Epsilon calculus in the sense of the addition of critical formulas with additional epsilon for universal quantifier is conservative for an intermediate logic if all classical quantifier shifts are available.

Remark

This provides a variant of the second epsilon theorem, however, not leading to the cut-free derivation of quantified formula. The same holds for Herbrand theorem.

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